Lab 1 – Wireshark Installation and Basic Packet Analysis

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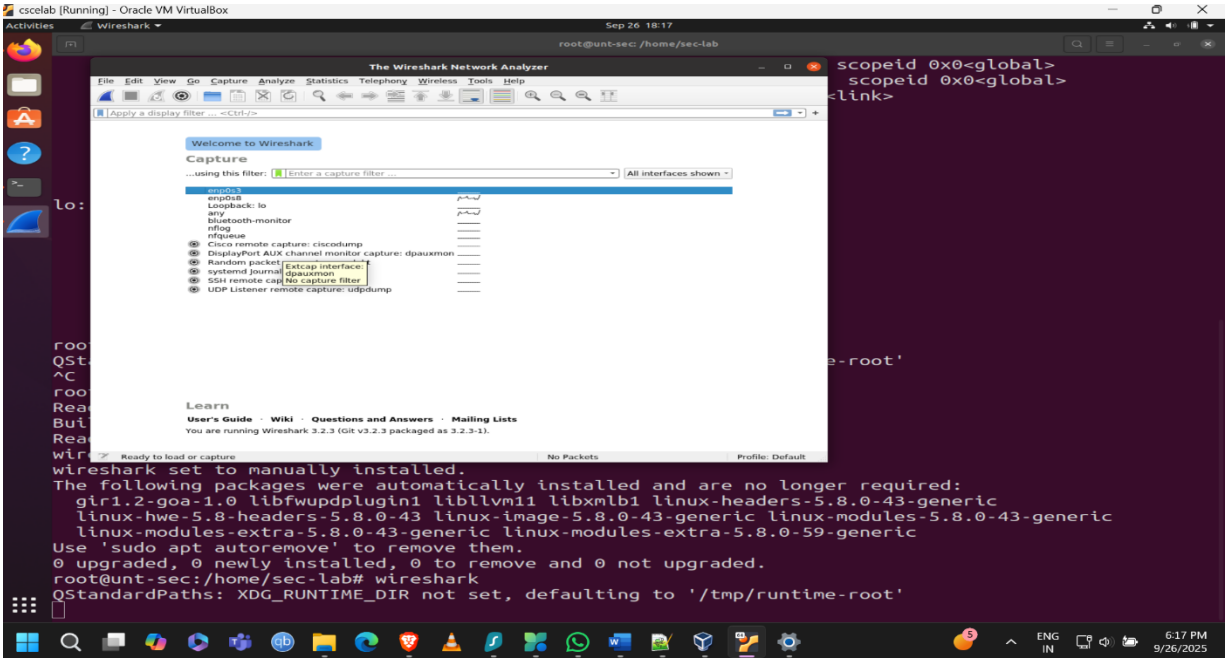
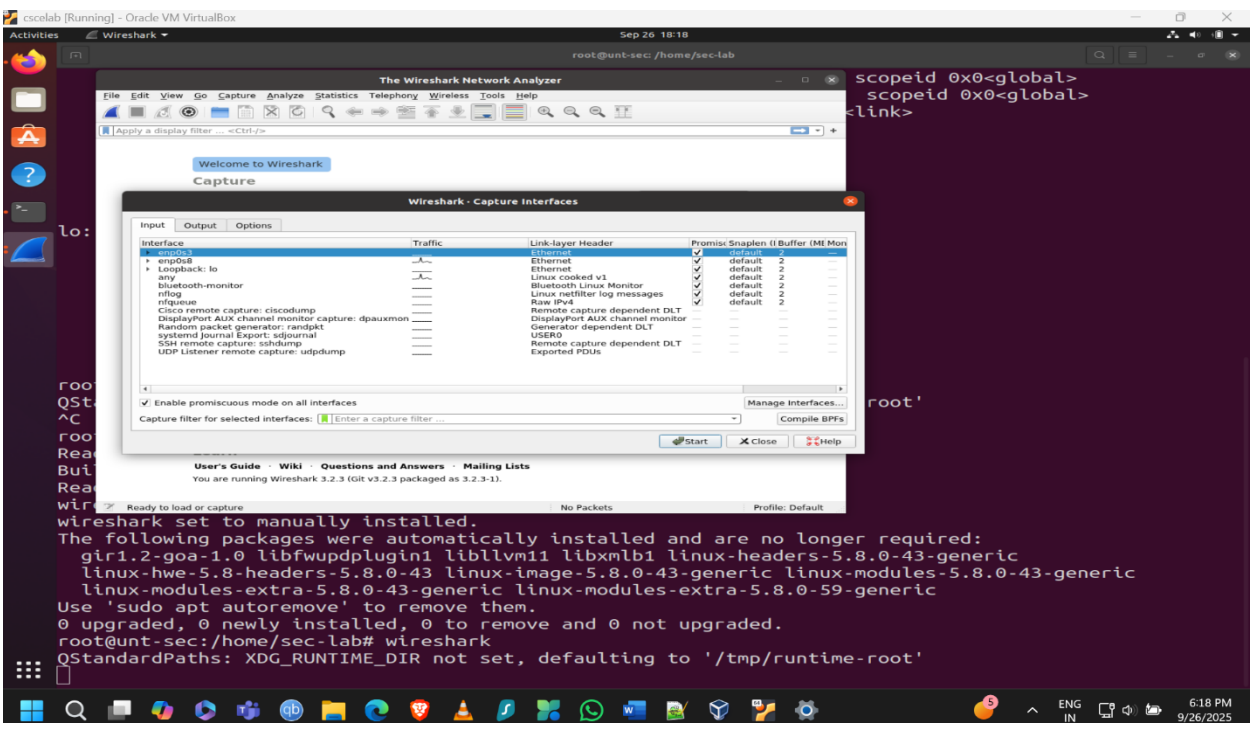
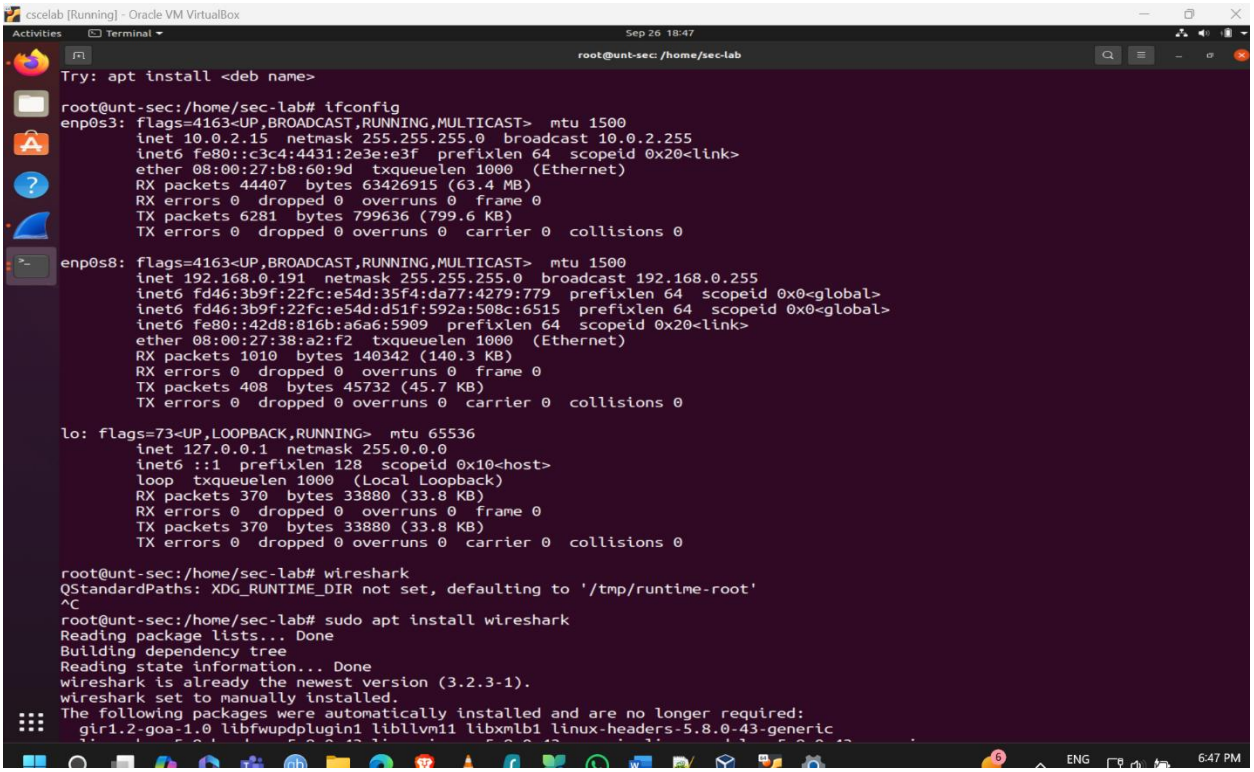
Installed Wireshark and running it.



Interfaces shown/ detected in the terminal



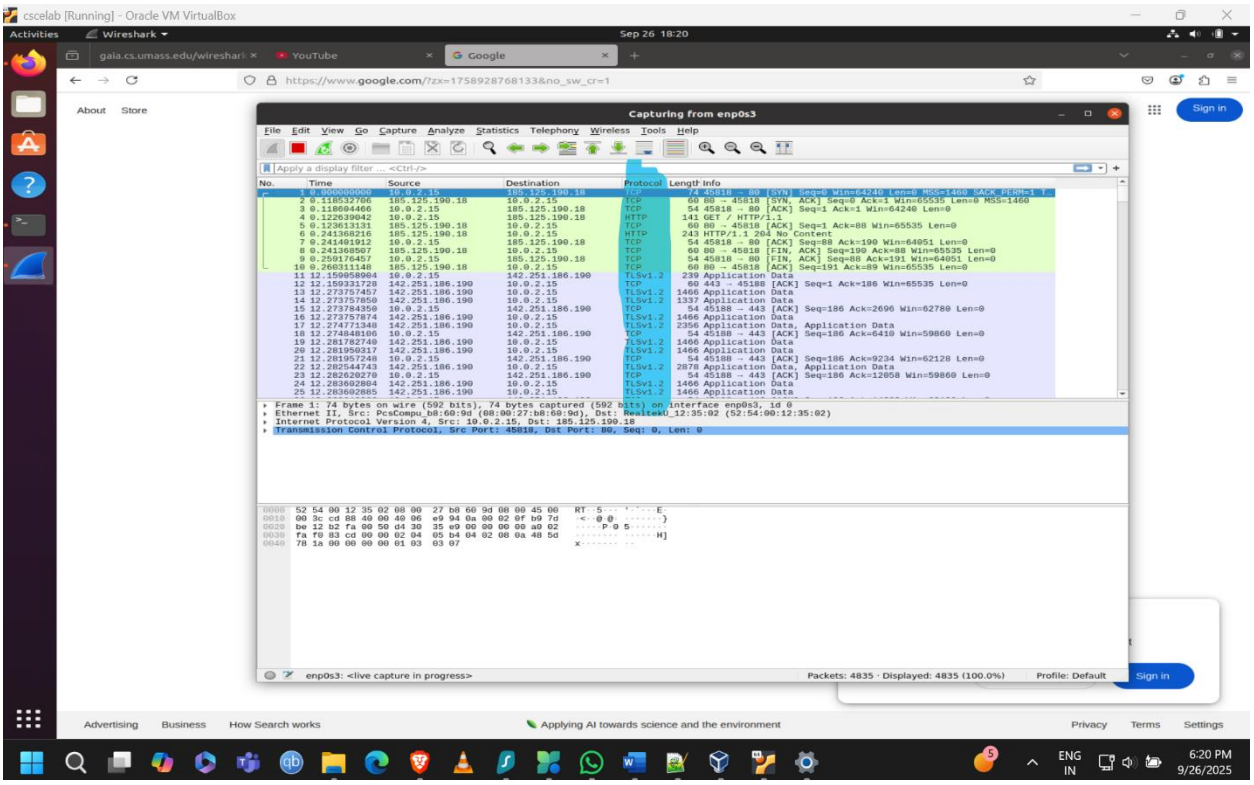
For reference my ifconfig network information is as follows



Starting capture on the ethernet we capture most of the packets passing through our network

We can see several protocols that runs to transmit these packets

We can clearly see the highlighted part it has protocols



TCP (Transmission Control Protocol)

TCP is a transport-layer protocol that provides a reliable, connection-oriented communication channel between two endpoints.

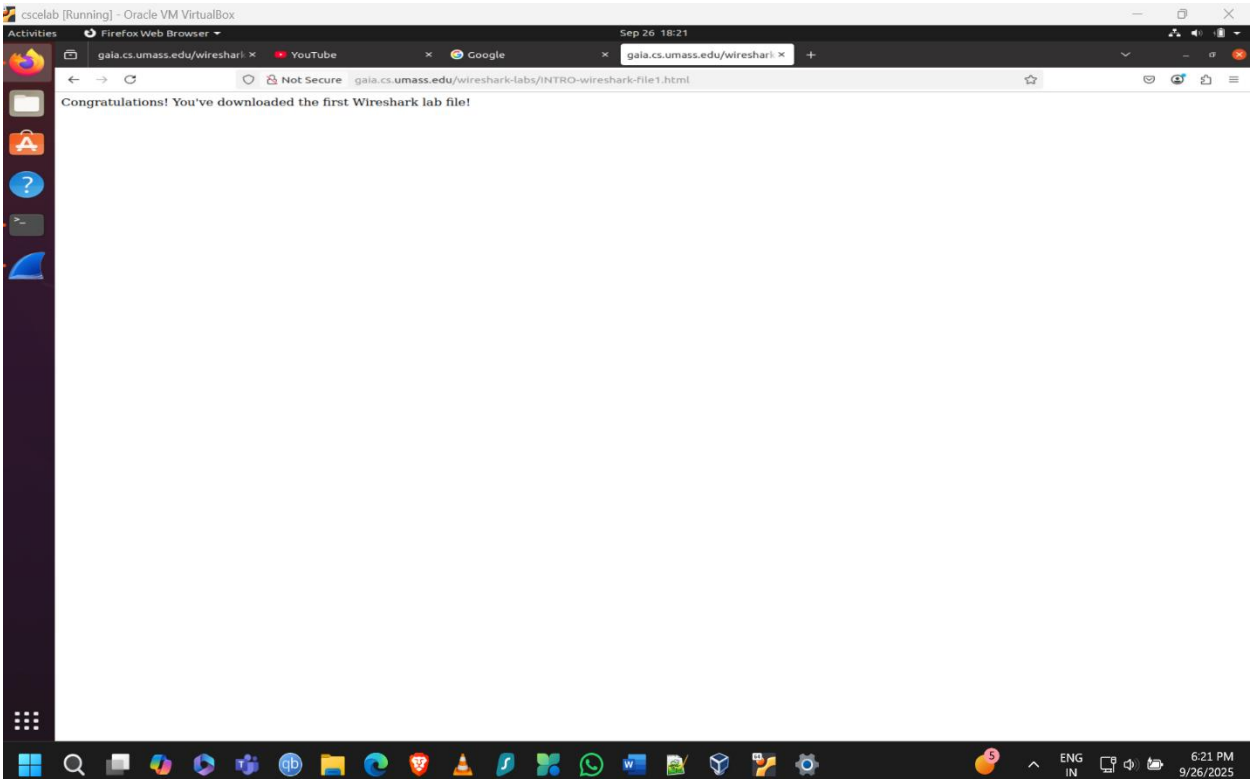
HTTP (Hypertext Transfer Protocol)

HTTP is an application-layer protocol used by web browsers and servers to exchange requests and responses for web resources (HTML pages, images, etc.). It mainly uses the client–server model in which client issues **GET** requests and the server replies with **status codes** like 200 OK 404 sever not found.

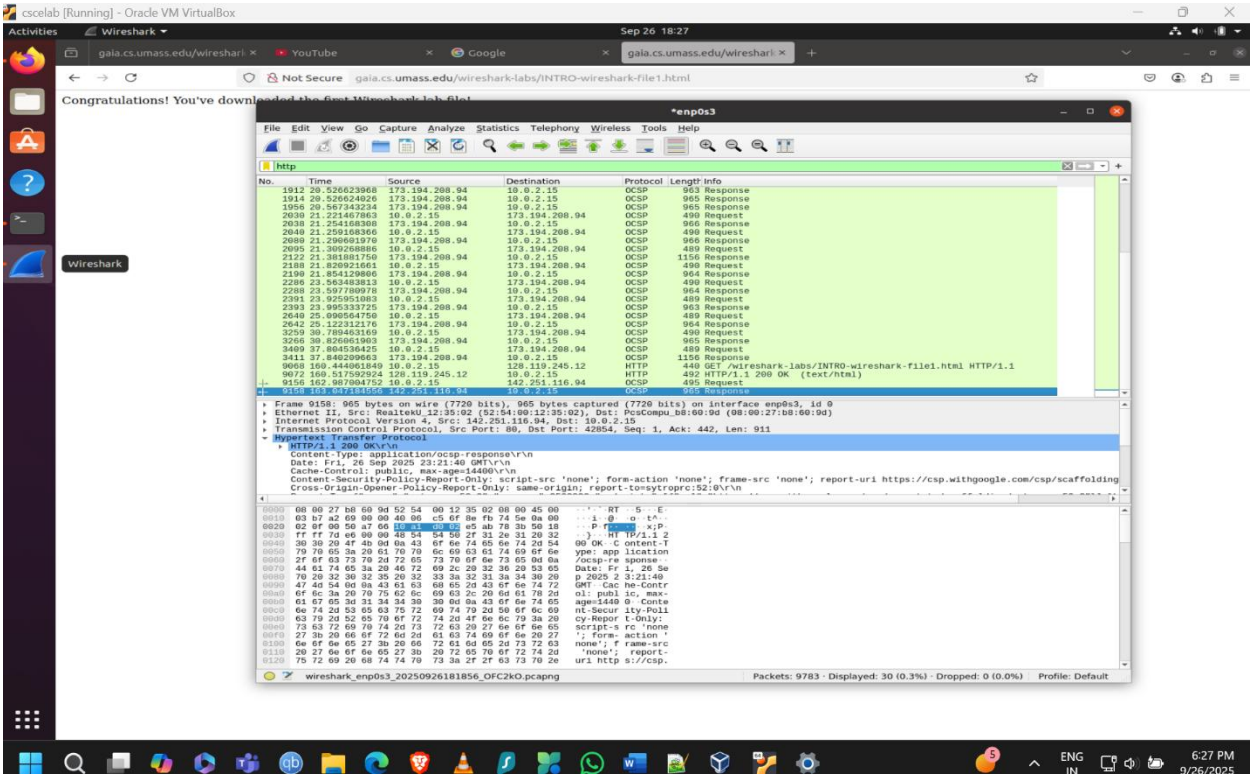
TLSv1.2 (Transport Layer Security, version 1.2)

TLS is a cryptographic protocol that secures communications over the network by providing **encryption**, **integrity**, and **authentication**. TLS 1.2 newer version and widely deployed to secure HTTPS traffic.

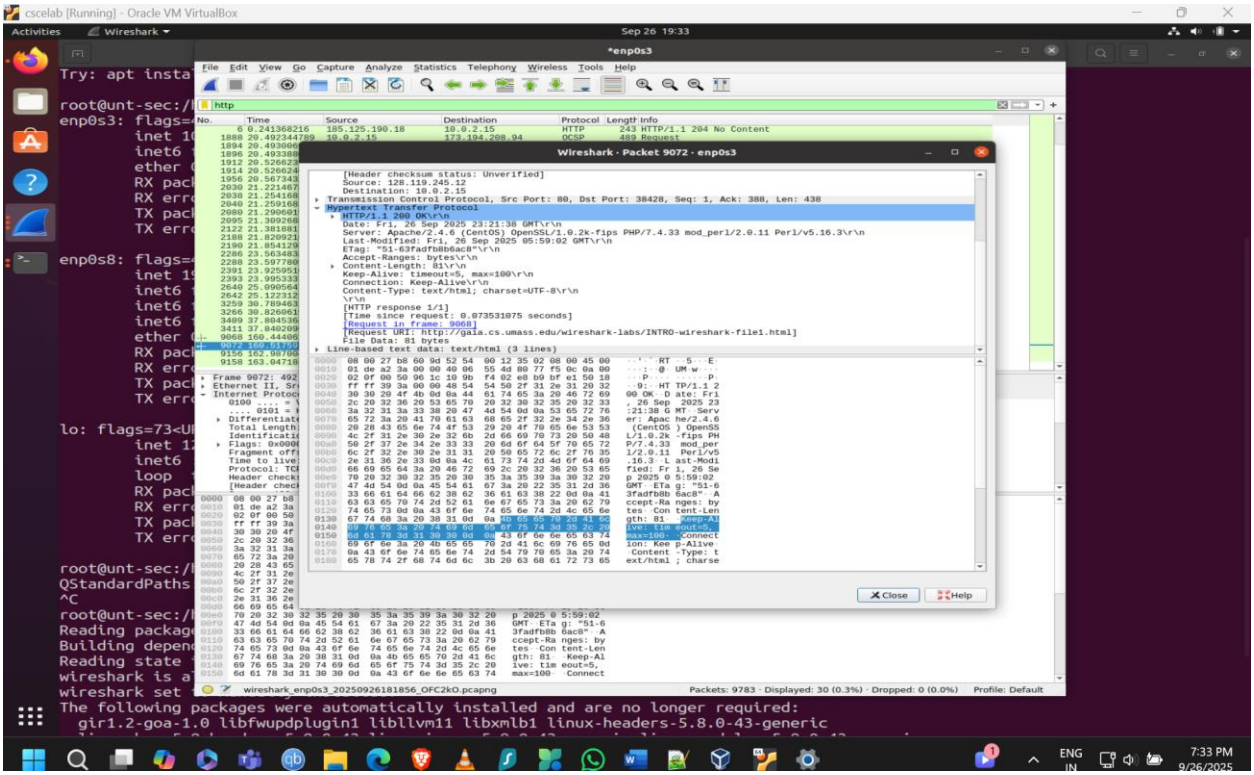
Visiting gaia http site mentioned in the lab manual



Filtering through the HTTP packets

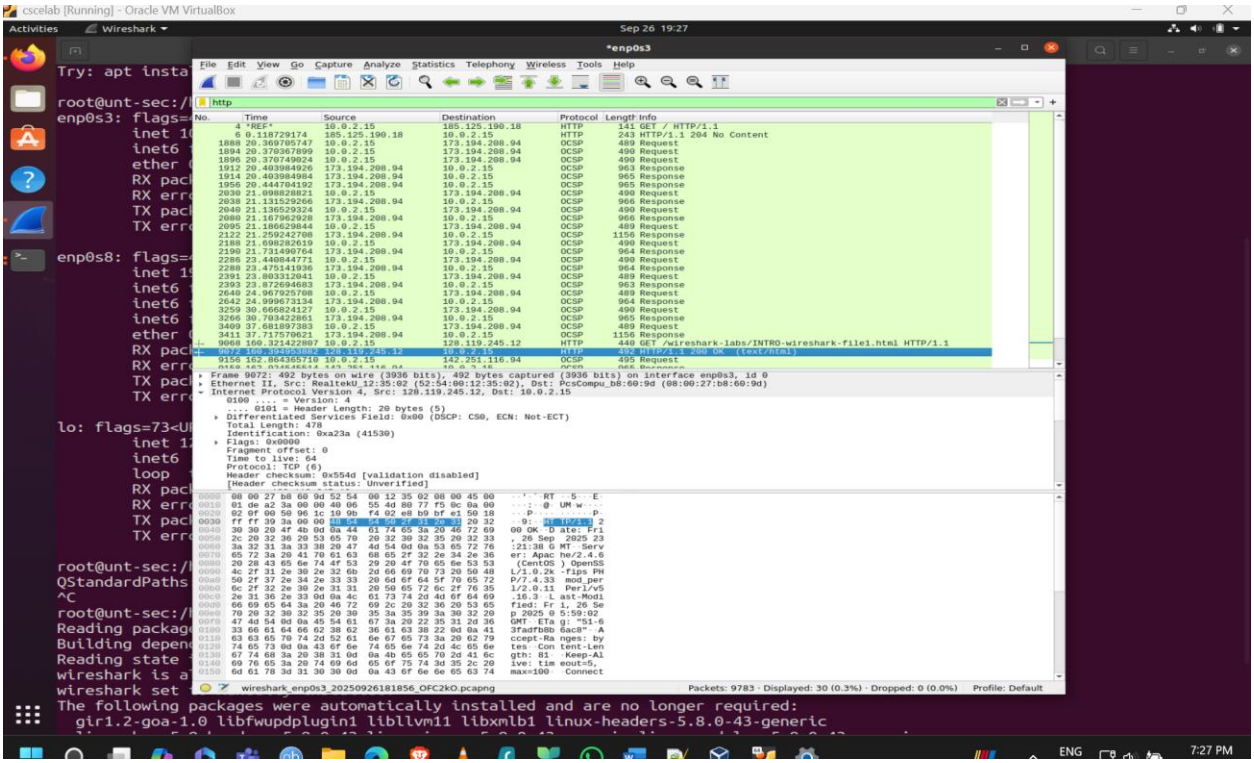


Inspecting HTTP get 200 ok packet

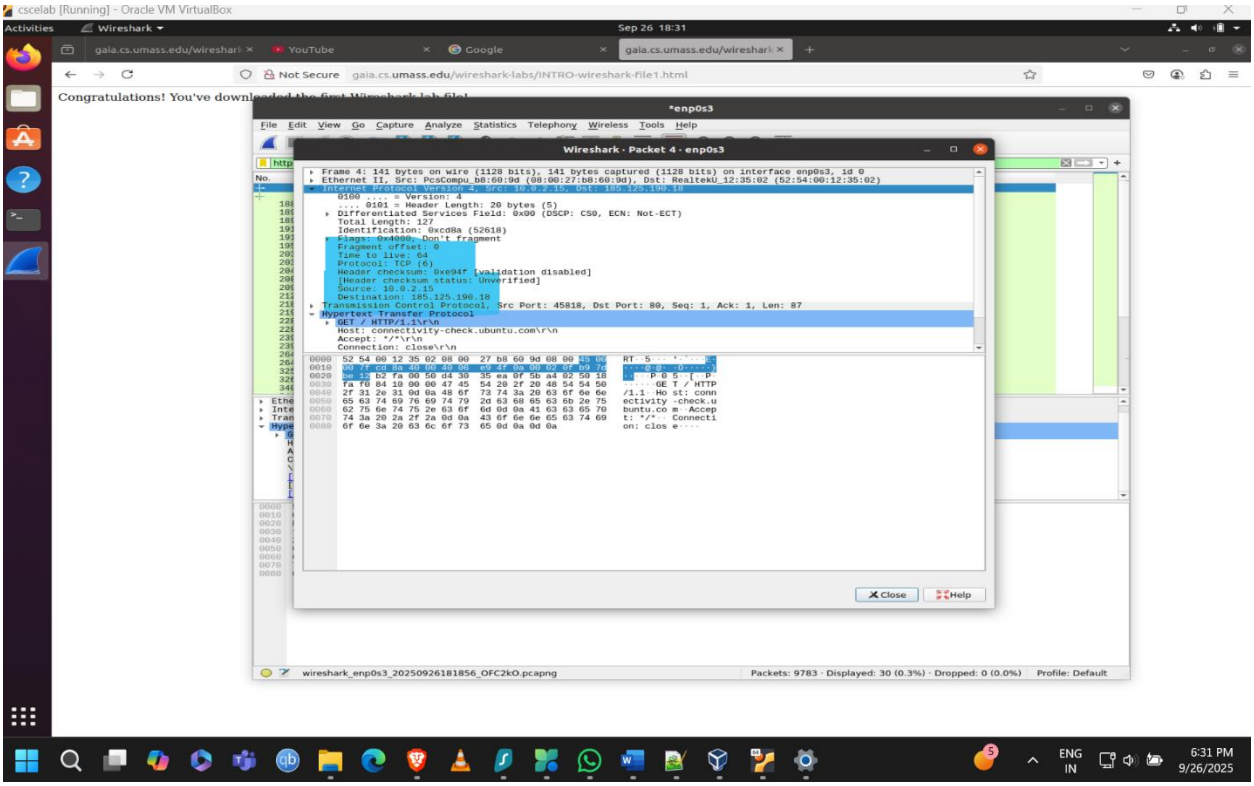


the time until HTTP GET OK reply we can get that by setting time reference feature in the options menu

Here we can clearly see that it took **160.394953882 seconds** for the server to send the 200 OK after the GET request.



Here, double clicking on the packet we can see the source and destination ip address as highlighted below

Source as : 10.0.2.15

Destination as : 185.125.190.18